DATA SCIENCE PROJECT

Telco Customer Churn Analysis

CRISP-DM Framework • Random Forest Classifier • Power BI Dashboards

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7,043	21	80%	0.84
Customers	Features	Accuracy	ROC-AUC

AGENDA

01 **Business Understanding**
Objectives, stakeholders & research questions

02 **Data Overview**
Dataset structure, features & quality

03 **Key EDA Findings**
Tenure, contracts, internet service & LTV

04 **Machine Learning Model**
Random Forest — training & evaluation

05 **Stakeholders**
Who benefits and how

06 **Recommendations**
Actionable business strategies

01 | Business Understanding

BUSINESS OBJECTIVES

1. Identify key drivers of customer churn
2. Build a predictive model to flag at-risk customers
3. Quantify financial impact via Lifetime Value (LTV)
4. Deliver actionable retention recommendations

RESEARCH QUESTIONS

RQ1 How does tenure differ for churned vs. retained customers?

RQ2 Which internet service type has the highest churn rate?

RQ3 Do Month-to-Month contracts drive higher churn?

RQ4 What is the LTV gap between churned & retained customers?

RQ5 Which variable most strongly influences LTV?

02 | Data Overview

7,043
Customers

21
Features

26.5%
Churn Rate

80%
Model Accuracy

Demographics

gender, SeniorCitizen, Partner, Dependents

Services

PhoneService, MultipleLines, InternetService, Streaming, Security, Backup...

Account Info

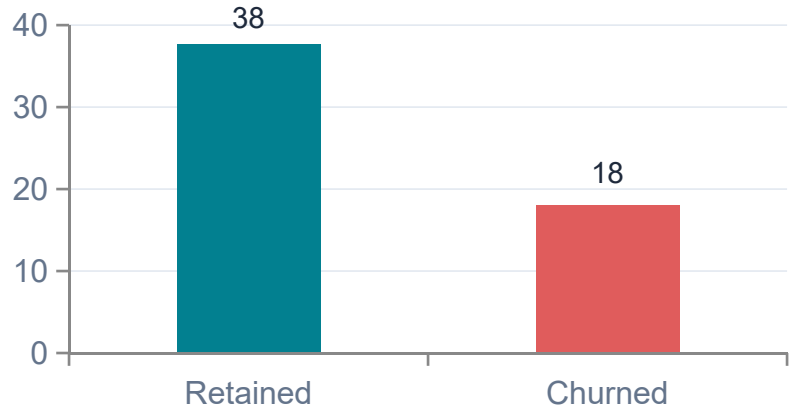
Contract, PaperlessBilling, PaymentMethod, tenure

Financials

MonthlyCharges, TotalCharges + engineered LTV

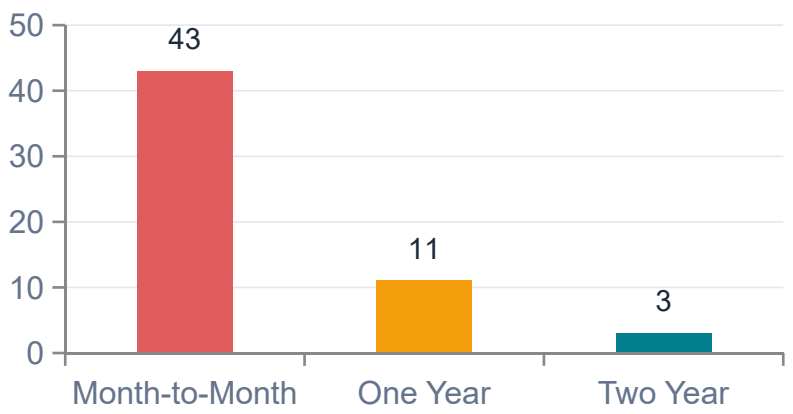
03 | Key EDA Findings — Tenure & Contract

RQ1 | Average Tenure by Churn Status



Retained customers have 52% more tenure on average. The first 12–24 months is the highest-risk churn window.

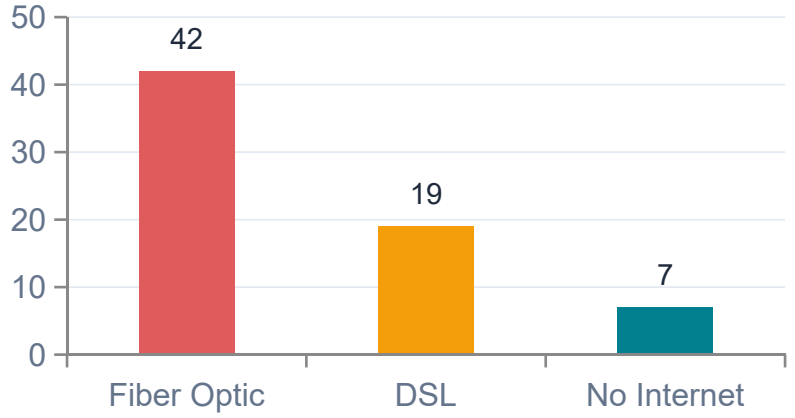
RQ3 | Churn Rate by Contract Type



Month-to-month customers churn at 9× the rate of two-year contract customers. Contract migration is the highest-ROI retention lever.

03 | Key EDA Findings — Internet Service & LTV

RQ2 | Churn Rate by Internet Service



Fiber optic customers churn at 42% — over 1.6× the overall average. Service quality or pricing issues likely responsible.

RQ4 | Lifetime Value — Churned vs Retained

Retained customers generate ~2× the Lifetime Value of churned ones. Most churners leave before generating significant revenue.

04 | Machine Learning Model

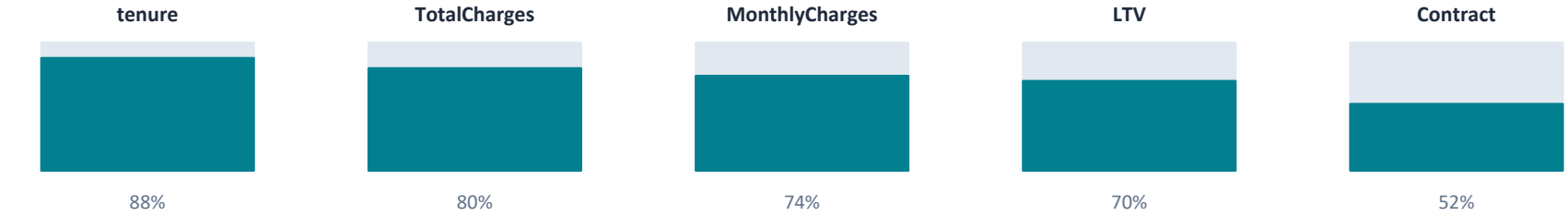
RANDOM FOREST — CONFIGURATION

Algorithm	Random Forest Classifier
Train/Test Split	80% / 20% (stratified)
n_estimators	100 trees
random_state	42 (reproducible)

MODEL PERFORMANCE METRICS

Metric	Retained	Churned	Overall
Overall Accuracy	—	80%	80%
Precision	83%	67%	—
Recall	92%	46%	—
F1-Score	87%	55%	—
ROC-AUC	—	—	0.84

TOP FEATURE IMPORTANCES



05 | Stakeholder Register

Chief Revenue Officer
HIGH

Executive Sponsor

Revenue impact, LTV, retention ROI

Marketing Manager
HIGH

Primary User

Segmentation, campaign targeting

Product Manager
HIGH

Primary User

Fiber quality, contract incentives

Data Science Team
HIGH

Model Owners

Model pipeline, retraining, AUC

CRM / Success Team
MED

Operational Users

Churn alerts, CRM integration

Finance Team
MED

Reviewer

Budget justification via LTV

Network / Tech Ops
MED

SME

Fiber root cause investigation

Customer Support
LOW

Data Contributors

Early churn signals from contacts

06 | Business Recommendations

01

Contract Migration Campaign

Offer month-to-month customers annual plan incentives. A 10–15% conversion could cut churn by ~4 percentage points.

02

Fiber Optic Retention Programme

Deploy NPS surveys for Fiber customers in months 1–6. Review pricing and assign dedicated technical support.

03

Deploy Model in Production

Score all active customers monthly. Trigger automated CRM workflows for High-Risk customers (probability > 60%).

04

Early Lifecycle Engagement

Build a structured onboarding journey with loyalty milestones for customers in their first 12 months.

05

LTV-Based Retention Spend

Prioritise budget for Fiber + Month-to-Month customers — highest LTV at risk. Retention spend is justified up to the LTV gap.

THANK YOU

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80%	0.84	~2×	9×
Accuracy	ROC-AUC	LTV Gap	Contract Risk